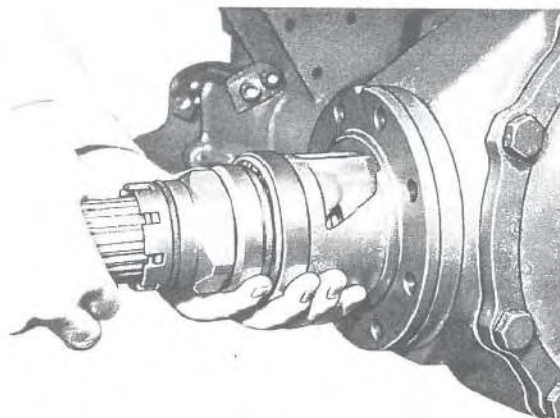


10 Coat pinion flange with sealing compound No. 1 and drive pinion bearing with running-in paste No. 3. Install with the shims determined on the data sheet.

Note: Install oil duct on pinion flange facing upwards. The O-ring between the pinion flange and drive housing has been eliminated.

11 Install wheel hub drive, refer to 6.1/1 or 6.2/1.



UR33-0562

12 Coat oil ring with lubricating grease and install in torque tube using special tool No. 10.



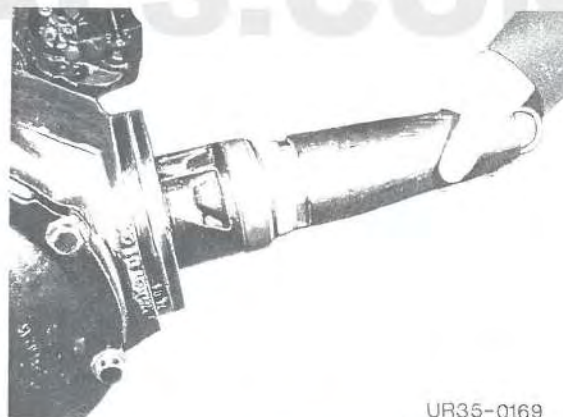
UR35-0270

13 Slip special tool No. 15 onto drive pinion, coat torque tube flange with sealing compound No. 1, bolt torque tube to pinion flange and tighten using special tool No. 2. For tightening torques, refer to 1.5/1.

14 Clean splines and cavity of propeller shaft, fill cavity behind the profile with lubricating grease No. 2 (50 g.) and install propeller shaft.

15 Attach spring brake cylinder, measure adjustment dimension from hole center of clevis to the hole center of actuating lever (lever must rest on the stop). If necessary adjust by shifting the spring brake cylinder in the oblong holes.

For adjustment dimensions, refer to 1.3/1.



UR35-0169

Note: Do not actuate emergency release device.

16 Coat fastening bolts with locking agent No. 4 and bolt on axle struts, tighten to 350 Nm.

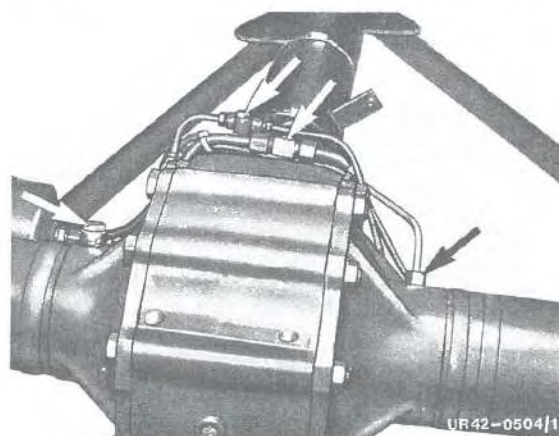
17 Attach brake lines, line for differential lock, axle breather, and spring brake cylinder.

18 Fit road wheels.
For tightening torques, refer to 1.5/1.

19 Remove axle from assembly fixture.

20 Install rear axle, refer to 2.1/1.

Note: Retighten wheel nuts after 50 km.

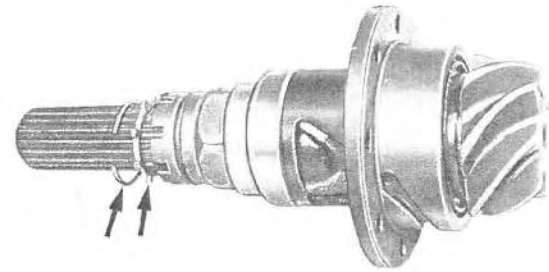


UR42-0504/1

Disassembly

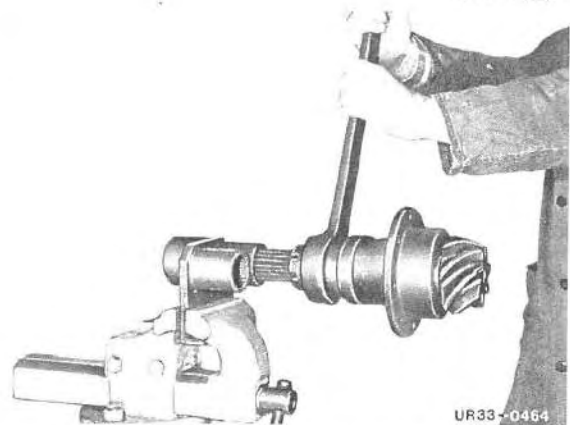
1 Remove pinion flange with drive pinion, referring to 2.1/1.

2 Remove lock washer, serrated washer and sealing ring from hexagon nut.



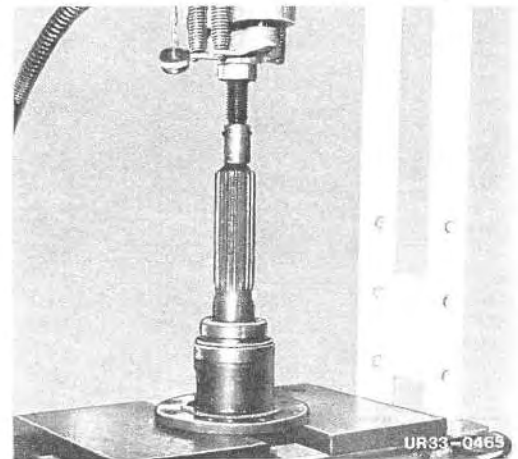
UR33-0463/1

3 Clamp drive pinion with special tool No. 20 and unscrew hexagon nut using special tool No. 16.



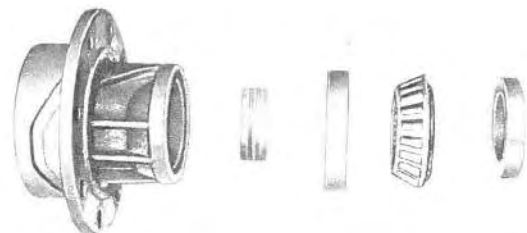
UR33-0464

4 Force drive pinion out.



UR33-0469

5 Remove retaining ring, taper roller bearing, bearing ring and clamping sleeve from pinion flange.

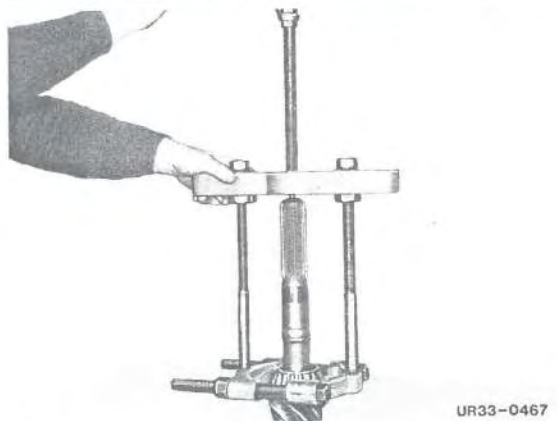


UR35-0221

747.111/113



6 Withdraw outer taper roller bearing race from pinion flange, using puller and special tool No. 9.



7 Draw taper roller bearing and inner race off drive pinion.

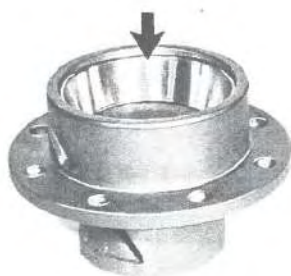
8 Clean and check all parts, exchange if necessary.

Note: Prior to installing taper roller bearing, rinse out and insert with SA 90 oil.



Important!

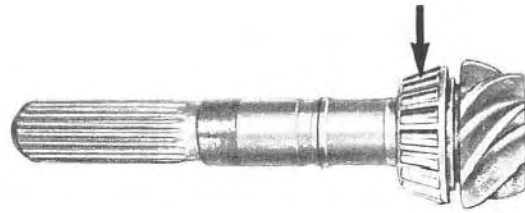
If drive pinion is to be exchanged, it is always to be replaced **together** with ring gear. Make sure serial Nos. are identical !



Assembly

1 Force outer taper roller bearing race into pinion flange.

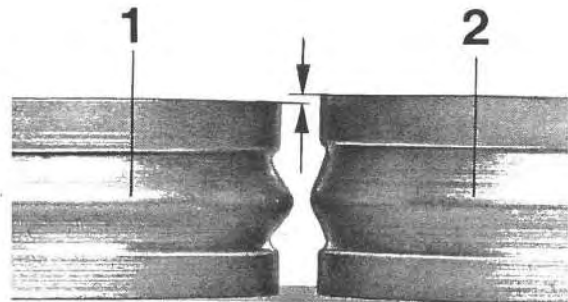
- 2 Press taper roller bearing onto drive pinion.
- 3 Introduce drive pinion into pinion flange.



UR33-0468/1

- 4 Slip new clamping sleeve onto drive pinion.

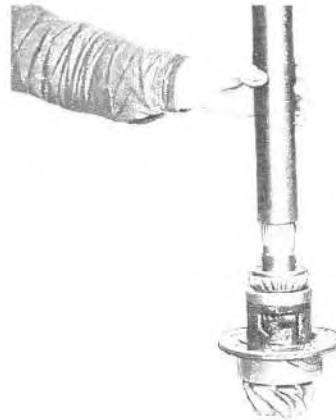
Note: Be sure to use **new clamping sleeve (2)** because old sleeve (1) is compressed from previous use.



UR 33-0009

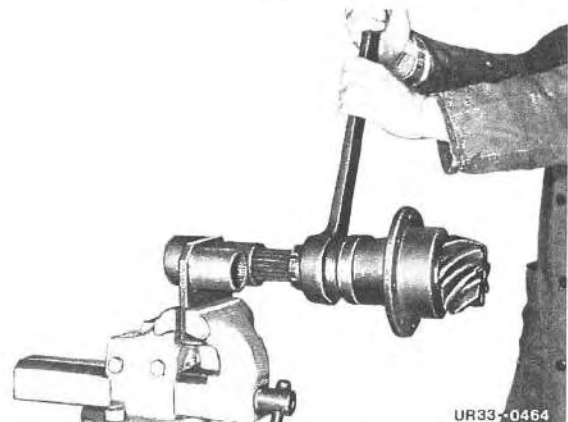
- 5 Coat taper roller bearing with lubricating grease No. 2. Press taper roller bearing into position so that it has 1 to 2 mm clearance.

- 6 Install retaining ring and screw hexagon nut on.



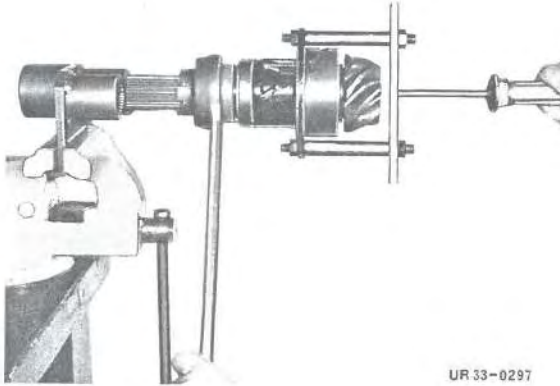
UR33-0495

- 7 Clamp drive pinion with special tool No. 20 and tighten hexagon nut with special tool No. 16 until all play disappears.



UR33-0464

747.111/113

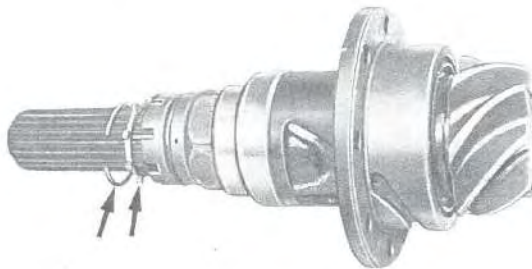


UR 33-0297

8 Apply special tools No. 17 and No. 18 to pinion flange and continue to torque hexagon nut with special tool No. 16 until break-away torque of 3 to 3,5 Nm is reached.

Note: Break-away torque is reached just before hexagon nut leaves its neutral position.

Drive pinion will have to be removed and operations 3 to 8 repeated if torque setting is exceeded.



UR33-0463/1

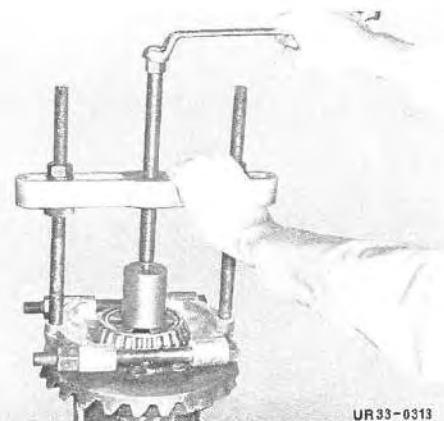
9 Fit serrated washer, lock washer and sealing ring in hexagon nut.

10 Coat pinion flange with paste No. 3 and install pinion flange with drive pinion.

MBMANUALS.COM

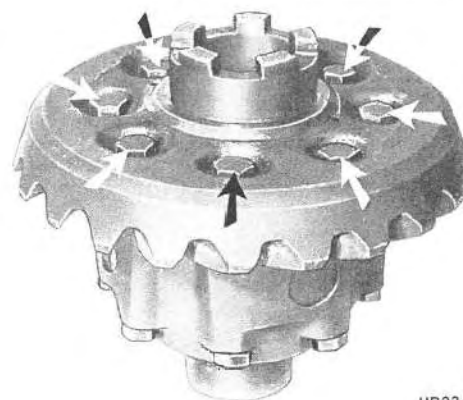
Disassembly

- 1 Remove rear axle, referring to 2.1/1.
- 2 Disassemble rear axle, referring to 3.1/1.
- 3 Pull both taper roller bearings off differential.



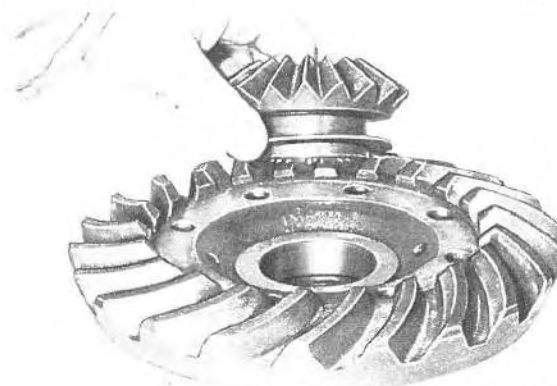
UR33-0313

- 4 Unscrew and remove ring gear from differential housing.



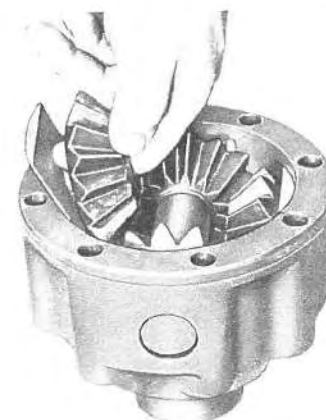
UR33-0469/1

- 5 Remove side gear and thrust washer.
- 6 Remove both short differential pinion shafts.



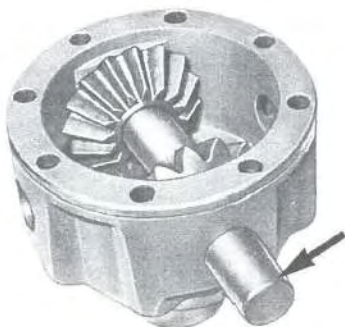
UR33-0470

- 7 Take thrust segments and differential pinions out.



UR33-0471

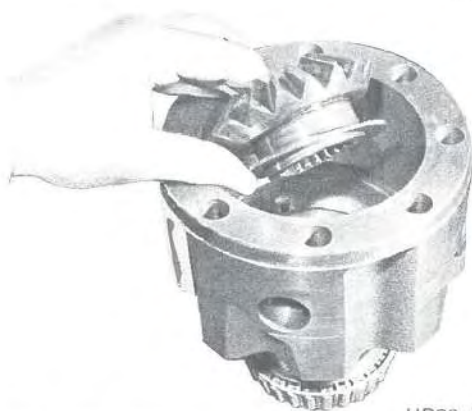
747.111/113



UR33-0472/1

8 Remove long differential pinion shaft.

9 Take thrust segments and differential pinions out.



UR33-0330

10 Take side gear and thrust washer out of differential housing.

11 Clean and check all parts, exchange if necessary.

MBMANUALS.COM

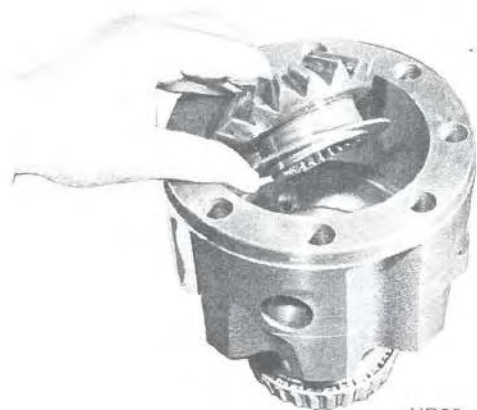
Assembly



UR33-0008

Note: Ring gear must only be exchanged together with drive pinion. Make sure serial numbers are identical.

1 Coat all differential pinion shafts and thrust segments with paste No. 3.



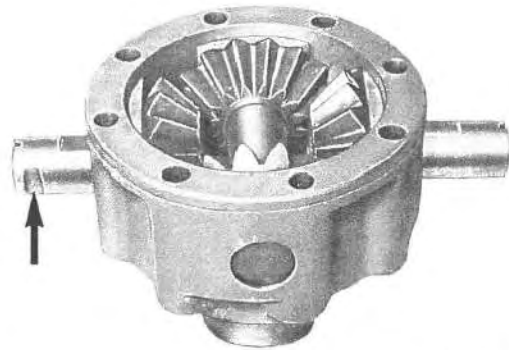
UR33-0330

2 Fit thrust washer and side gear in differential housing.

3 Fit thrust segments, differential pinions and differential pinion shafts.

Note: Note recesses in differential pinion shafts for bolts.

4 Insert thrust washer and side gear in differential.



UR33-0473/1

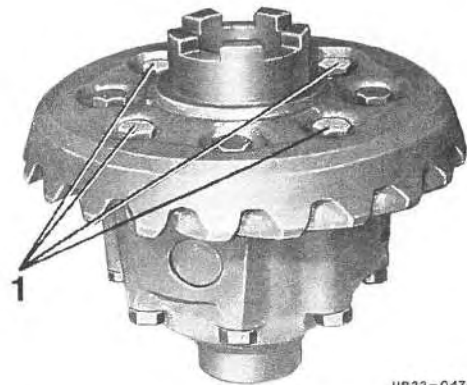
5 Position ring gear on differential housing and torque fastening bolts.
See 1.5/1 for tightening torques.

Note:

From Model	Chassis end No.	Rear axle No.
435.111	001 522	

Body-fit holes (1) with colour dot (white) are marked and reamed with reamer 13 H 7, install body-fit bolts only here.

Ream holes (1) of differential housing without marking with reamer 13 H 7 as before.



UR33-0474/1

6 Using special tool No. 22, press large inner taper roller bearing race onto differential housing and coat with fluid grease No. 5.

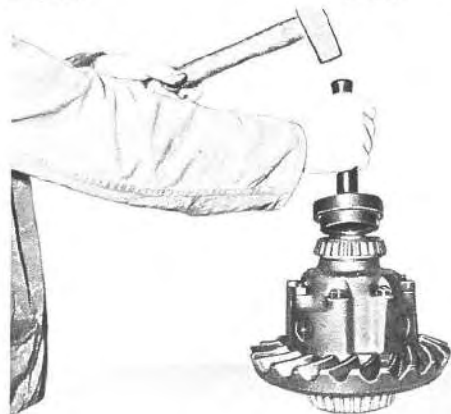


UR33-0475

7 Using special tool No. 22, press small inner taper roller bearing race onto differential housing and coat with fluid grease No. 5.

8 Assemble rear axle, referring to 3.1/1.

9 Install rear axle, referring to 2.1/1.



UR33-0476

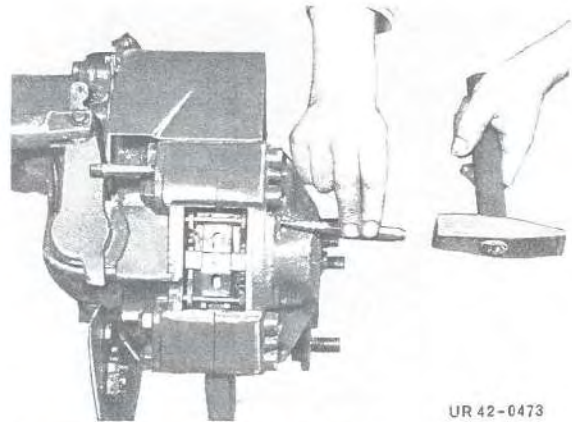
Disassembly

Note: Operations at left and right-hand wheel hub drives are to be executed in same order. Work is same for axle in situ.

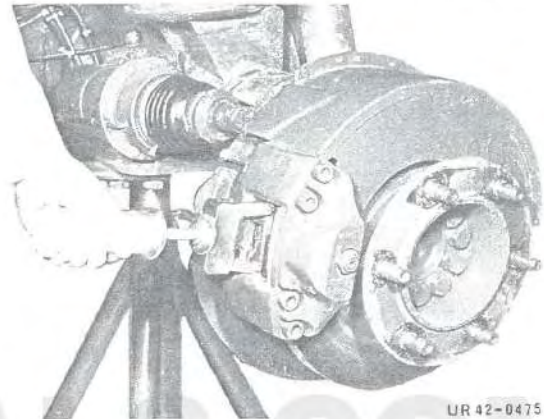
- 1 Remove rear axle, referring to 2.1/1.
- 2 Clean rear axle, clamp in trestle and detach wheels.
- 3 Detach metal cover at fixed caliper, knock pins out and remove spring clip.

Note: Detach brake pad wear indicator at left-hand fixed caliper.

- 4 Remove brake pad using special tool No. 13.



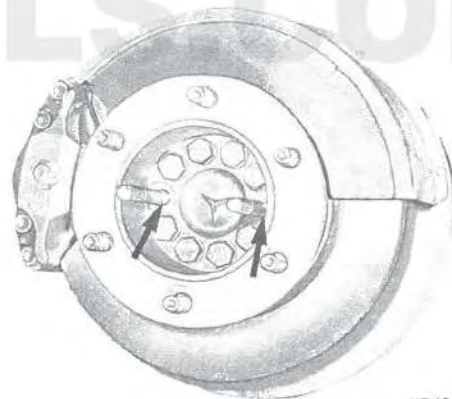
UR 42-0473



UR 42-0475

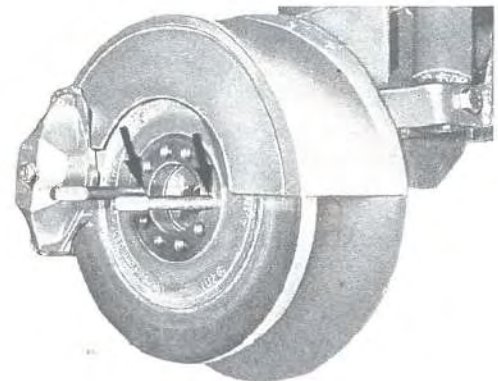
- 5 Unscrew two wheel hub fastening bolts, using special tool No. 4.

- 6 Insert special tool No. 5 and unscrew all other fastening bolts. Remove wheel hub.



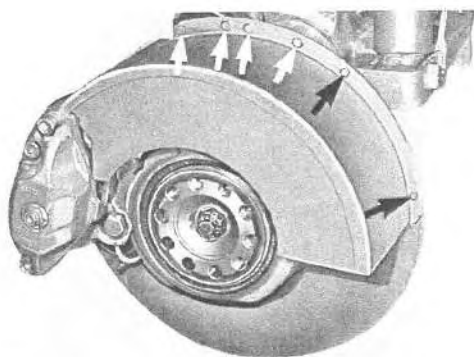
UR 42-0496/1

- 7 Unscrew special tool No. 5 and remove with brake disk.



UR 42-0497/1

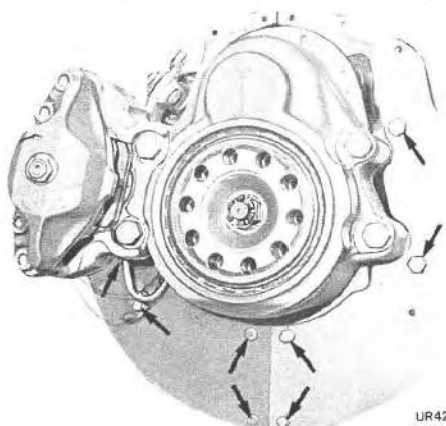
747.111/113



8 Detach brake backplate and guard plates.

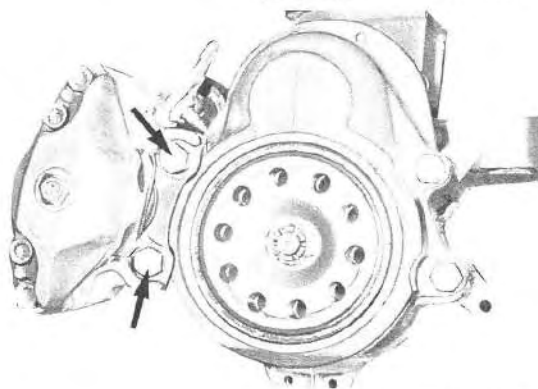
9 Release and remove pin from clevis of spring-loaded cylinder.

UR42-0498/1



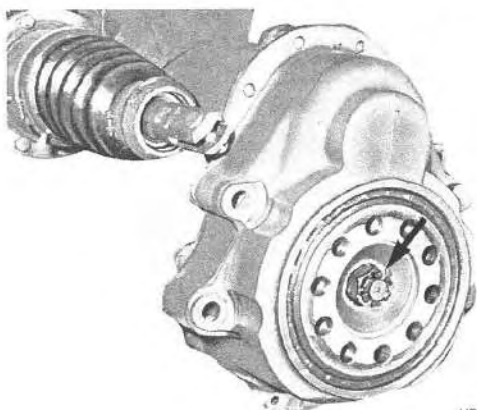
10 Unscrew brake line, wear indicator cable plug and metal covers.

UR42-0499/1



11 Unscrew fixed caliper.

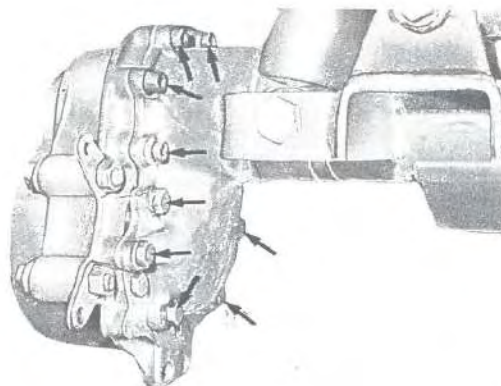
UR35-0276/2



12 Release and unscrew castle nut on clamping bolt.

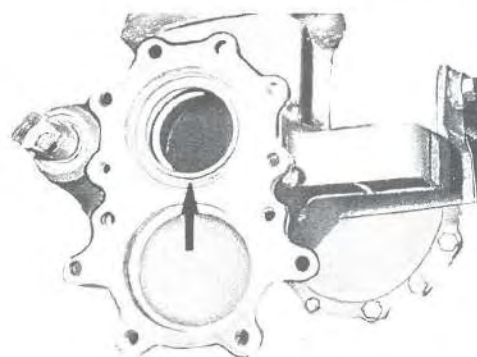
UR35-0274

13 Detach wheel hub drive, unscrew socket head bolts using special tool No. 2.



UR35-0275/1

14 Remove shaft sealing ring in the axle tube.



UR35-0277/1

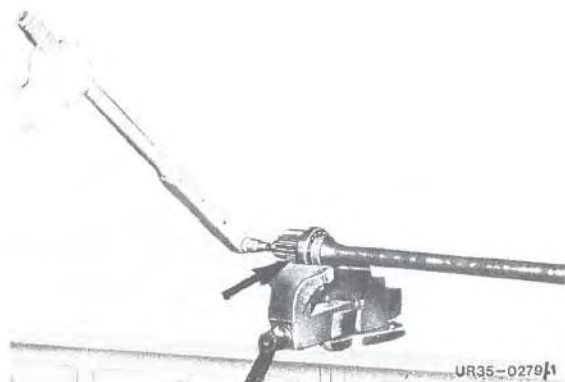
15 Remove rear axle shaft from the wheel hub drive.



UR35-0278/1

16 Clamp drive gear together with special tool No. 7 and unscrew center fastening bolt using special tool No. 3.

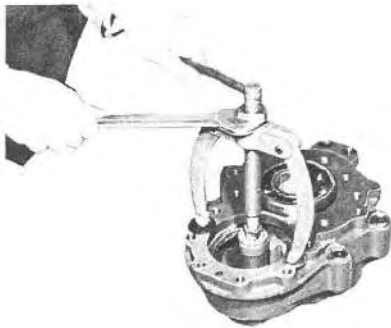
17 Check grooved ball bearing on drive gear for wear and renew if necessary.



UR35-0279/1

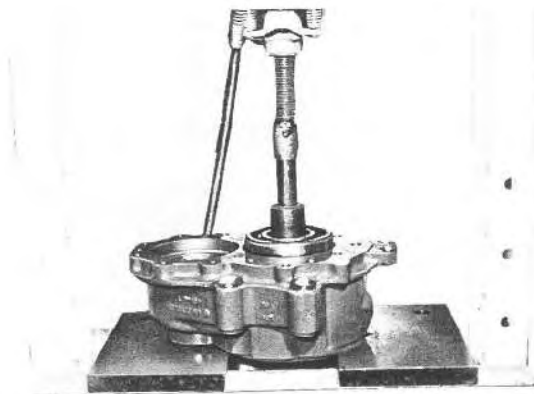
35.6 Disassembly and Assembly of Wheel Hub Drive

747.111/113



18 Check small roller bearing in the wheel hub drive housing for wear and renew if necessary.

UR35-0266



UR35-0265

19 Press driven gear together with roller bearing and shaft sealing ring out of the wheel hub drive housing.

Note: Slip assembly ring onto roller bearing so that rollers cannot fall out.

20 Remove bearing in the wheel hub drive housing.

21 Clean, check and renew all parts as required.



UR35-0264

Assembly

Note: In converting to the reinforced wheel bearing (SA 36 184) the work necessary in the area of the driven gear is carried out in accordance with 33.6 - 6.1/7.

Note: Prior to assembly determine difference between bearing seat in the axle tube and wheel hub drive housing, as well as thickness of the grooved ball bearings and pack with shims.

1 If the small roller bearing in the wheel hub drive housing is defective, press in new bearing coat liquid grease No. 5, using special tool No. 23.

Note: In wheel hub drive housing part No. 416 331 0405 the bearing seat must be packed with 1 washer 0,3 mm and 5 washers each 0,5 mm.

2 Install driven gear in wheel hub drive housing and press in deep - groove ball bearing. Fit sealing ring on deep - groove ball bearing.

3 Fit cylindrical roller bearing in wheel hub drive housing using special tool No. 12.

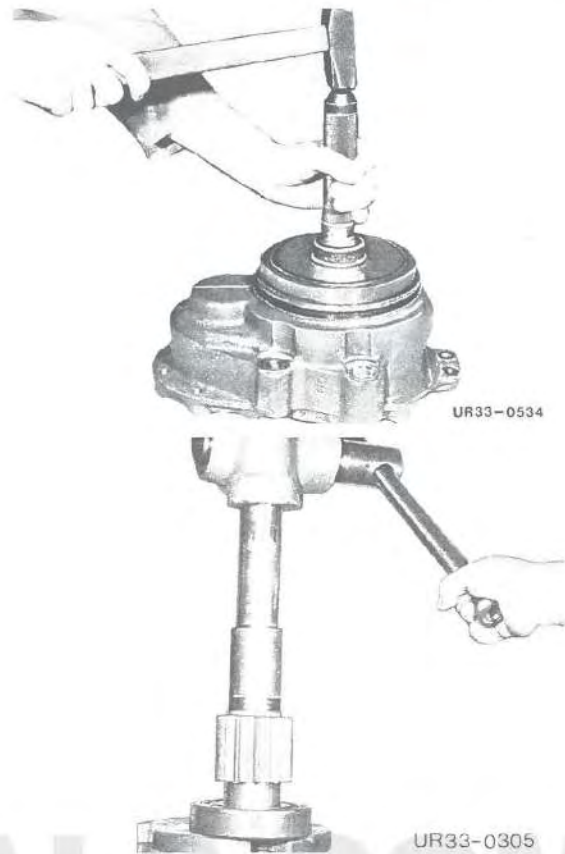
Note: Remove assembly ring from cylindrical roller bearing only when pressing in the bearing.



UR33-0531

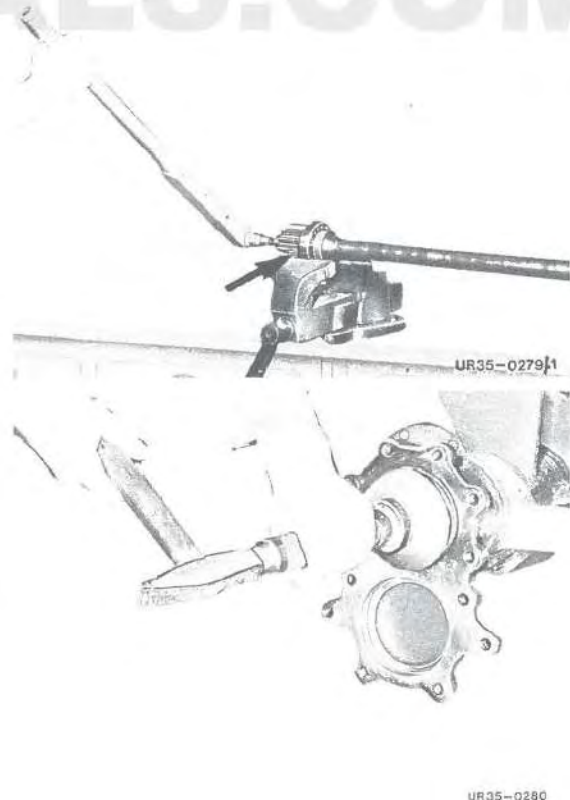
4 Coat shaft seal with lubricating grease No. 2, press onto wear ring and install using special tool No. 12.

5 Insert clamping bolt and screw castle nut on.



6 Press deep-groove ball bearing onto drive gear.

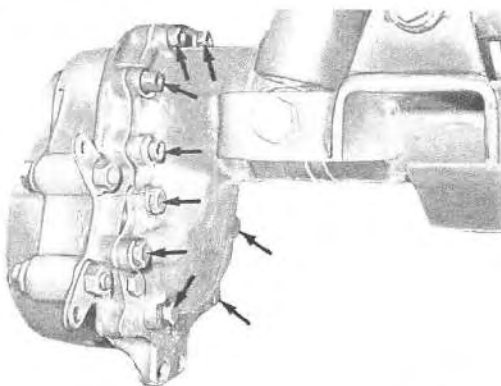
7 Clamp drive gear with special tool No. 7 and bolt drive gear to rear axle shaft with special tool No. 3 so that teeth mesh, then torque. See 1.5/1 for tightening torques.



8 Coat shaft seal with lubricating grease No. 2 and install in axle tube using special tool No. 11.

9 Coat splining of rear axle shaft with paste No. 3 prior to insertion. Using plastic-headed hammer, drive deep-groove ball bearing into bearing seat.

747.111/113

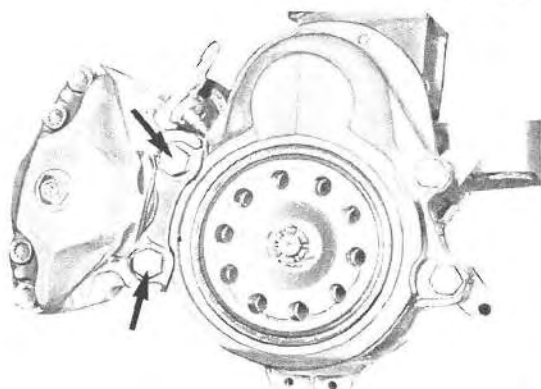


10 Coat contact surface of axle tube and fastening bolts with sealant No. 1. Attach wheel hub drive housing and tighten socket-head bolts with special tool No. 2.

Note: Coat rollers of small roller bearing with multipurpose grease.

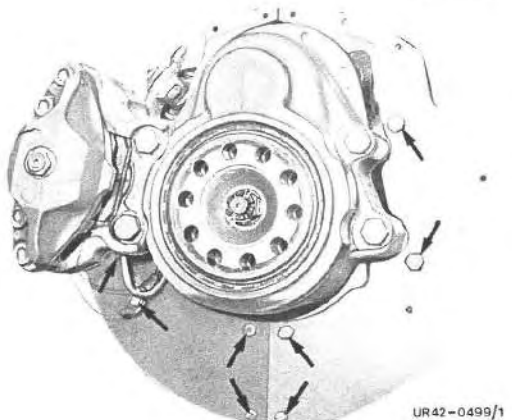
11 Torque clamping bolt castle nut and then secure.
See 1.5/1 for tightening torques.

UR35-0275/1



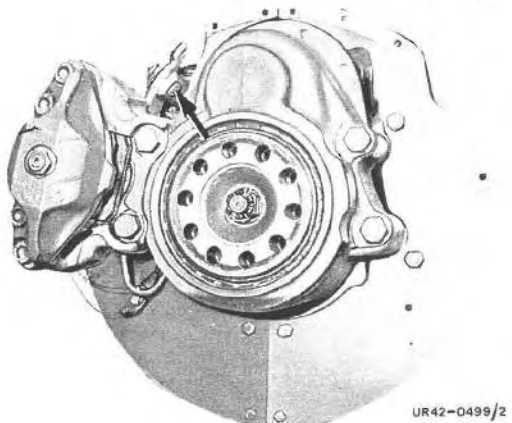
12 Torque fixed caliper. Insert body-fit bolt at side where brake disk emerges (straight-ahead position).
See 1.5/1 for tightening torques.

UR35-0276/2



13 Attach metal covers, cable plug for wear indicator, and brake line.

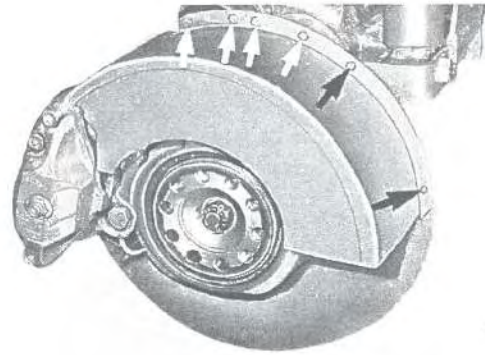
UR42-0499/1



14 Coat split pin in clevis of spring-loaded cylinder with paste No. 3 prior to inserting and securing. Check adjustment of quick release, correcting if necessary.
See 1.3/1 for adjustment.

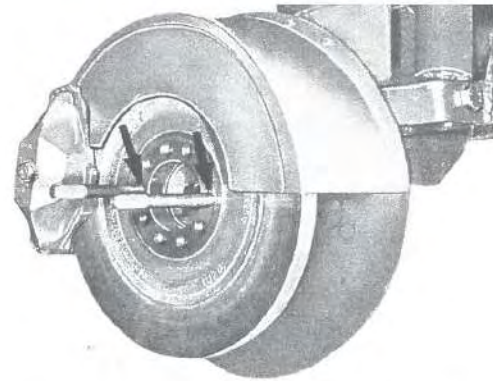
UR42-0499/2

- 15 Attach guard plates and brake backplate.



UR42-0498/1

- 16 Bolt special tool No. 5 in position and attach brake disk.



UR42-0497/1

- 17 Attach wheel hub and torque fastening bolts, using special tool No. 4.
See 1.5/1 for tightening torques.

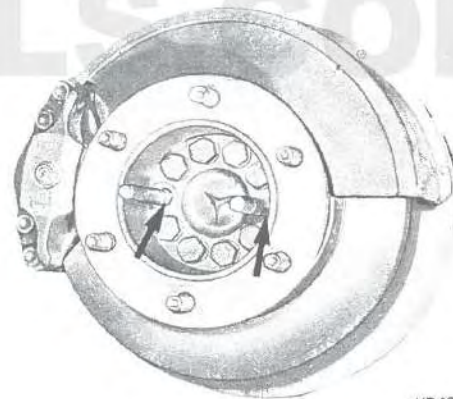
- 18 Install brake pads.
See 1.3/1 for permissible clearance.

- 19 Attach brake line.

- 20 Attach and torque wheels.
See 1.5/1 for tightening torques.

Note: Retorque wheel nuts after 50 km.

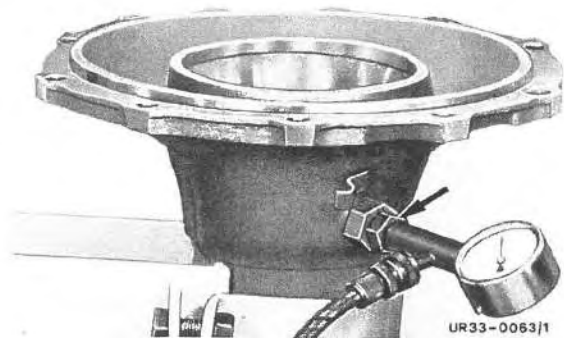
- 21 Install rear axle, referring to 2.1/1.
22 Bleed brake system and check for leakage.



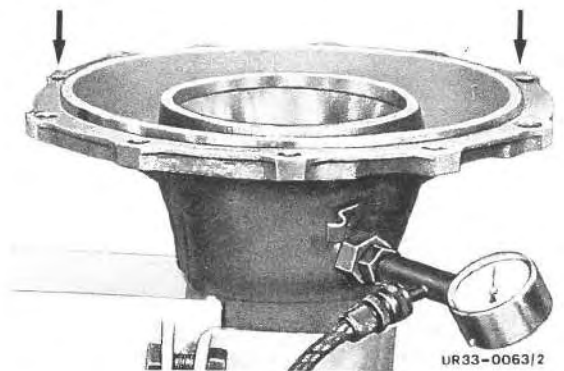
UR42-0495/1

Checking

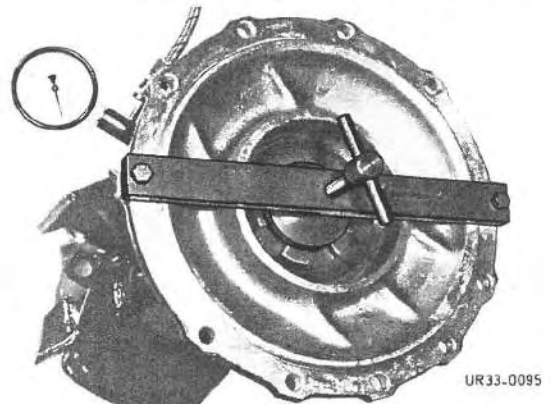
- 1 Connect pressure gage to differential lock.



- 2 Bolt special tool No. 19 to axle tube flange and apply pressure plate via spindle. Unscrew spindle by about 10 mm (this corresponds to position where differential lock is applied).

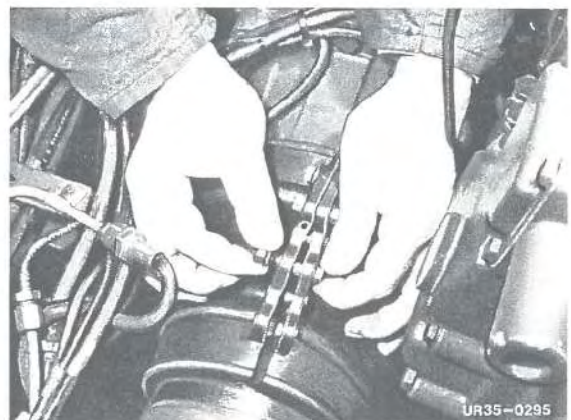
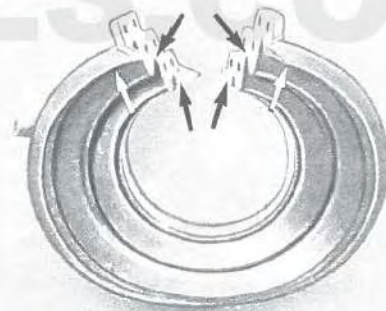
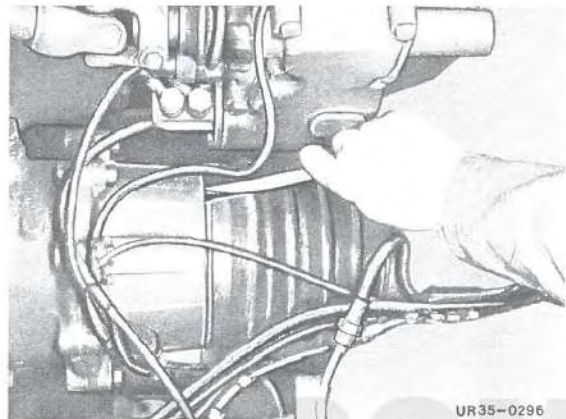
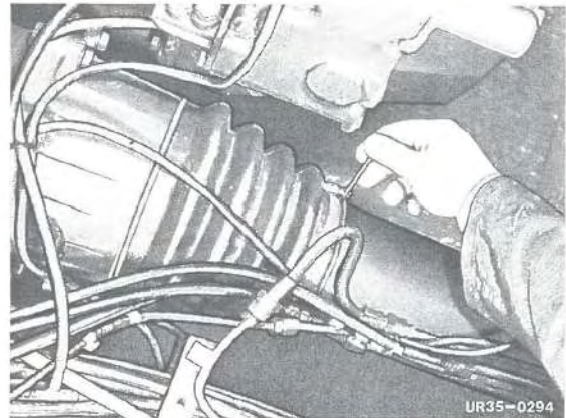


- 3 Connect air hose to pressure gage and admit 3 to 5 bar to piston chamber. Pressure must be maintained for several minutes after air supply is stopped.

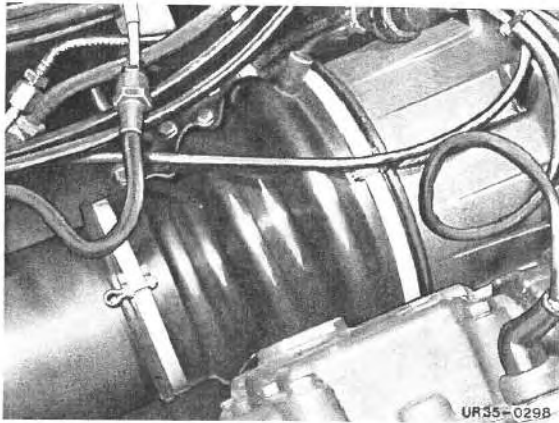


Exchange

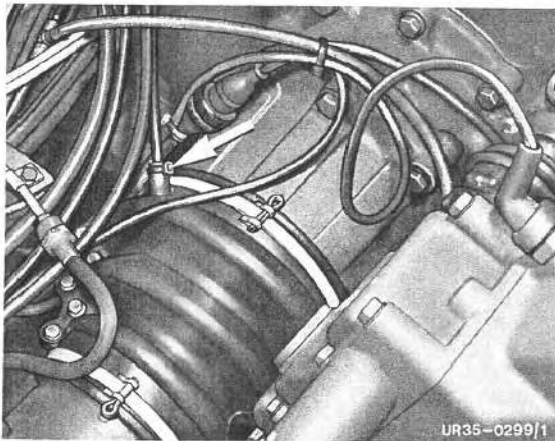
- 1 Remove spare wheel.
- 2 Remove hose clamps.
- 3 Cut rubber gaiter and take it off.
- 4 Clean sealing surfaces for thrust tube and torque ball casing.
- 5 Coat separating surface and ca. 2 cm of diameter sealing surface at junction with sealant No. 9.
- 6 Attach rubber gaiter to thrust tube and torque ball casing. Install eyelet with bolt and screw it with washer and nut on.



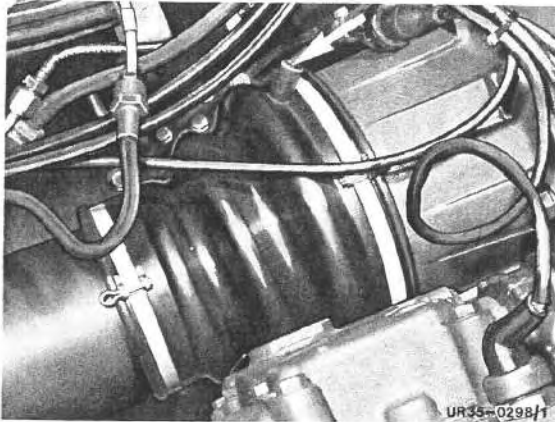
747.111/113



7 Turn rubber gaiter in the way that separating surface on left-hand side points horizontal to thrust tube. Then fit hose clamps.



8 Installing SA 35 977 (fording capacity), introduce exhausting line into fitting and fasten with hose clamp.



9 If SA 35 977 (fording capacity) is not to be installed, lock fitting with plug.

10 Install spare wheel.

Contents

Chapter	Page
1 General	
Tire pressures on fitting	1.1/1
Tire pressure table	1.1/2
Notes on correct tire fitting	1.1/2
Tires and track	1.2/1

MBMANUALS.COM

Tire fitting

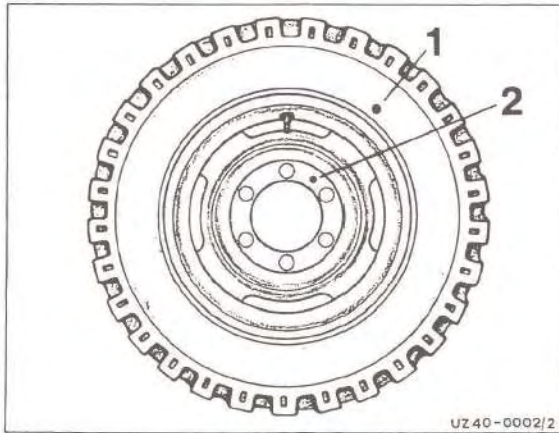
Observe the following measures to prevent tire fitting accidents:

1. Check wheels, rims, tubes and tires for evident damage.
2. Note safe fitting pressure, see table.
3. Use protective equipment.
4. Note general accident prevention regulations.

Safe tire fitting pressures

Group	Tire size	PR	Fitting pressure in bar
1	12.4 – 28	8	3.0
	12.4 R 28	8	
	14.9 – 24	8	
	14.9 R 24	8	
	14.9/80 – 24	10	
	14.9 – 30	10	
	18.4 – 30	10	
	18.4 R 30	10	
	16.9 – 30	12	
	16.9 R 30	12	
2	7.5 – 18	6	4.0
	10.5 – 18	8	
	12.5 – 18	8/10	
	7.5 – 20	8/12	
	10.5 – 20	6/8/10/14	
	10.5 R 20	8/10/12	
	12.5 – 20	10	
	14.5 – 20	10	
	14.5 R 20	10	
	9.5 – 32	10	
	9.5 – 36	12	
	12.4 – 38	12	
	22 – 20	10	
3	12.5 – 20	12	6.0
	12.5 R 20	12	
	14.5 – 20	18	
	14.5 R 20	18	
	14.5 – 24	16	
	14.5 R 24	16	
	16/70 – 24	14	
	9.00 – 20	14	
	9.00 R 20	14	
	11.00 – 20	16	
	11.00 R 20	16	
	12.00 – 20	18	
	12.00 R 20	18	
	12.00 R 22.5	—	
	13.00 R 22.5	—	
	20–20	16	

The air pressure used for tire fitting may exceed the manufacturer's operating tire pressure by up to 50% but must not amount to more than 4.5 bar, even if tires have higher PR rating and operating pressure. It is imperative for safety reasons that the bead is correctly seated on the rim at a pressure not higher than 3 bar.



Notes on correct tire fitting:

Tires with red marking

1 Tire is correctly seated when red mark on tire wall (1) agrees with center punch mark (2) in disk wheel.

Note: The red mark may consist of one or two dots.

Meanings:

Two red dots: Static unbalance below 0,4 % at lightest point of tire.

One red dot: Permissible static unbalance above 0,4 % at lightest point of tire.

Two center punch marks: Radial runout of disk wheel 0 to 0,5 mm.

One center punch mark: Radial runout of disk wheel up to 1,7 mm.

Tires with green marking

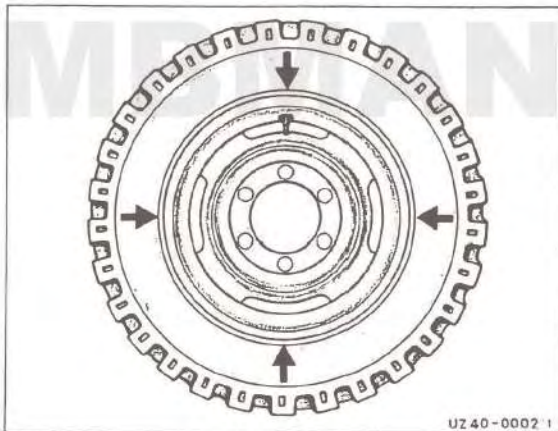
1 Tire is correctly seated when green mark on tire wall (1) agrees with center punch mark (2) in disk wheel.

Note: Green mark represents point of minimum radial force fluctuation.

2 Inflate tires to prescribed fitting pressure so that tire bead contacts wheel flange correctly. Check spacing of marking line. This must be even around the entire wheel circumference.

Note:

1. Use only one type of tread at any on axle. It is advisable to use the same type throughout.
2. Retreads are acceptable at the rear axle only.
3. Caution — do not balance wheels with lead weights attached.



Any tire that cannot be seated satisfactorily at the specified fitting pressures will have to be turned around on the wheel.

Inflate tire again to specified fitting pressure. See tire pressure table for safe pressure after fitting.

Tire pressure table ¹⁾

Chassis Model	Tires	Disk wheel	PR	Tread	At permissible axle weight rating kp	Tire pressure in bar	
						Front axle	Rear axle
435.115 .117	9,00 – 20	7 x 20	14	2)	4000	5,5	6,5
	9,00 R 20					5,5	6,5
	10,5 – 20 ³⁾	9 x 20	10	MPT	4000	4,75	4,75
	12,5 – 20	11 x 20	10	MPT	3700/4000	3,5	4,0
	12,5 R 20		12			4,0	4,0
	14,5 – 20	11 x 20	10	MPT	3960	2,5	3,0
	14,5 R 20		10/40		3960/4000	3,0	3,5
435.110 .111 .113	11,0 – 20	8,5 x 20	16	2)	5300	6,0	6,5
	11,00 R 20		16			6,3	7,5
	12,00 – 20	8,5 x 20	18	2)	5300	5,5	6,5
	12,00 R 20		18			5,8	7,0
	12 R 22,5	9 x 22,5		2)	5300	6,3	7,5
	13 R 22,5					5,8	7,0
	14,5 – 20	11 – 20	18	MPT	5290	4,5	5,0
	14,5 R 20		18			5,0	5,5
	14,5 – 24	11 x 24	16	MPT	5300	3,8	4,0
	14,5 R 24		16			4,0	4,5
	16/70 – 24	13 x 24	14	MPT	5300	3,0	3,5
	13,00 R 20	10,00V – 20			5300	5,0	5,0
	14,75/80R20						

1) Be sure to observe specified tire pressures on-highway

2) Truck tires when on-highway service predominates

MPT = Multipurpose tire for on and off-highway

Tires and track

Chassis Model	Tires	Disk wheel	Dish mm	Tire dia. mm	Tire width mm	Track mm
435.115 .117	9,00 – 20	7 x 20	158	1012	276	1745
	9,00 R 20			1018	258	
	10,5 – 20	9 x 20	65	955	298	1931
	12,5 – 20	11 x 20	100	1040	351	1860
	12,5 R 20			1030		
	14,5 – 20	11 x 20	100	1095	383	1860
	14,5 R 20					
435.110 .111 .113	11,00 – 20	8,5 x 20	160	1080	314	1890
	11,00 R 20				301	
	12 R 22,5	9 x 22,5	184	1084	310	1840
	12,00 – 20	8,5 x 20	160	1120	337	1890
	12,00 R 20			1122	322	
	13 R 22,5	9 x 22,5	184	1125	330	1840
	14,5 – 20	11 x 20	184	1095	383	1840
	14,5 R 20					
	14,5 – 24	11 x 24	184	1195	383	1840
	14,5 R 24					
	16/70 – 24	13 x 24	184	1175	442	1840
	13,00 R 20	10,00V – 20	184	1144	370	1840
	14,75/80 R 20					

Survey

Version

Brake System 435.115/117

42.11

Brake System 435.110/111/113/160

42.14

MBMANUALS.COM

Contents

Chapter	Page
1 General	
Installation Survey	1.1/1
Technical Data	1.1/1
Special Tools	1.2/1
Tightening Torques	1.3/1
Necessary Materials	1.3/1
Sectional View, Fixed Caliper	1.4/1
Direction of Rotation Specifications for Adjusting Combined Fixed Calipers	1.5/1
Troubleshooting Hydraulic Brake System	1.6/1
Troubleshooting Wheel Brakes	1.7/1
Brakes Functional Diagram	1.8/1
Exploded View	1.9/1
2. Fixed Caliper on Front Axle	
Removal and Installation of Brake Pads in Front Axle Fixed Caliper	2.1/1
Removal and Installation of Sealing Rings in Front Axle Fixed Caliper	2.2/1
3. Fixed Caliper on Rear Axle	
Removal and Installation of Brake Pads on Rear Axle Fixed Caliper	3.1/1
Disassembly and Assembly of Rear Axle Fixed Caliper	3.2/1
Adjusting Clearance for Rear Axle Fixed Caliper	3.3/1
4. Checking and Adjusting ALB Modulator	
Checking and Adjusting Failsafe Facility	4.1/1
Checking and Adjusting Load-Dependent Control	4.2/1
5. Bleeding Brake System	5.1/1
6. Mechanical Brake Pad Wear Measurement	6.1/1

Installation Survey

Chassis Model	Sales designation	Axles Model	Brake system ¹⁾ Tractor
435.115	U 1300 L	737.111 747.111	Dual-circuit hydraulic system and dual-circuit air brake system
435.117	U 1300 L/ 37	737.111 747.113	Disk brake ALB at rear axle ²⁾ Spring parking brake
			At permissible gross vehicle weight $\geq 7,5$ t Second pair of fixed caliper at rear axle ³⁾ (SA 35 988)

1) For pneumatic section, refer to group 43.11 - 1.1/1

2) ALB at rear axle and half front axle to chassis end No.

435.115: 084 388

435.117: 071 795

3) For functional diagram, refer to group 43.11 - 1.7/15

Technical Data

Front and Rear Axle

Brake Disk

			SA 35 988
Outside diameter	mm	374,00 373,90	370,00 369,50
Inside diameter	mm		80,40 80,30
Thickness	mm		22,0 21,9
Permissible wear limit	mm		20,5
Permissible runout	mm		0,1

Brake Pads

Pad size when new	mm	10,0
Minimum pad size	mm	2,0
Pad, quality		Agid 504 Energit 394

Fixed caliper

Clearance	mm	0,15 – 0,20 ¹⁾
-----------	----	---------------------------

1) 0,3 mm for model 435.115 with SA 35 988 (service brake with greater clearance)

Special Tools

Cons. No.	Number	Tool	Tool set
1	Test gauge for brake linings	440 589 00 19 00	B
2	Tester	001 589 64 21 00	B
3	Piston sealing plate	425 589 00 25 00	B
4	Twin hose for bleeder	000 589 02 25 00	B
5	Angled connection for bleeder	000 589 03 25 00	B
6	Counter support for extracting brake pads with special tool No. 7	000 589 34 33 00	B
7	Extractor for brake pads	001 589 43 33 99	B
8	Holder for installer	000 589 00 40 00	B
9	Installer for piston on front axle	406 589 07 43 00	B
10	Installer for protective caps at rear axle	406 589 08 43 00	B
11	Installer for protective caps at front axle	406 589 09 43 00	B
12	Piston resetting tool with tools No. 8 and No. 9 or No. 10 and No.11	000 589 52 43 00	B
13	14 mm index socket	001 589 70 09 00	C
14	Remover for compression spring	406 589 04 14 00	C
15	Mounting fixture for fixed caliper	406 589 04 31 00	C
16	Installation drift for end plate	406 589 10 43 00	C
17	Installation drift for sealing ring	406 589 11 43 00	C

Tightening Torques

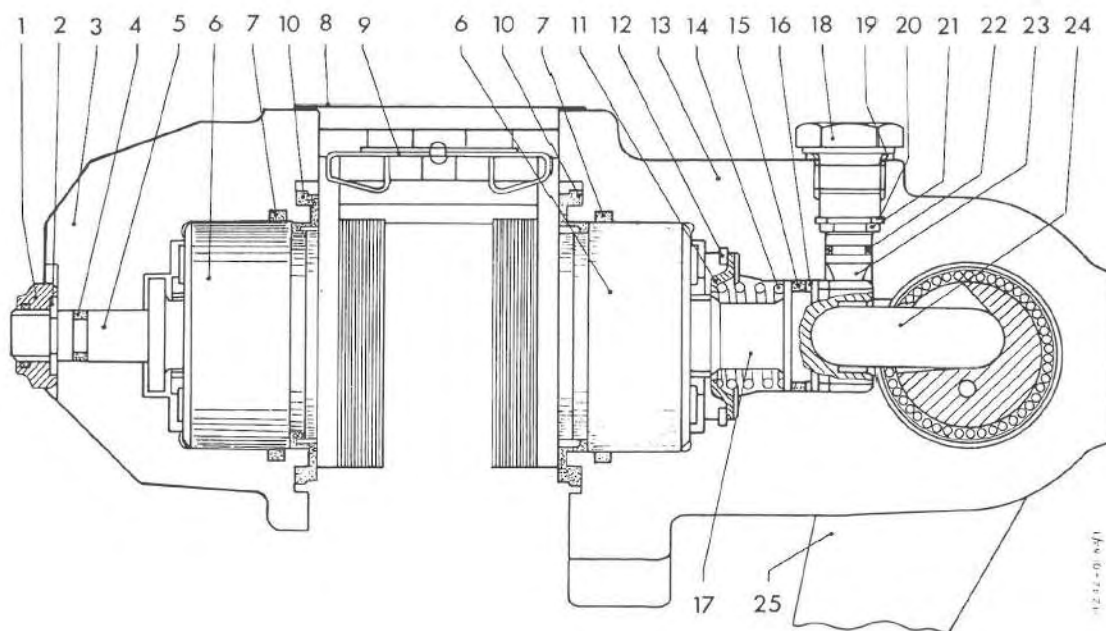
Designation		Thread	Nm
Wheel nuts		M 20 x 1,5	400
Fixed caliper to wheel hub drive housing		M 16 x 1,5	250
Fixed caliper	Counternut (cover side)	M 14 x 1	40
	Screw plug (flange side)	M 16 x 1,5	35 ⁻⁷
	Bleeder valve		4 ⁻²
	Necked-down bolts		280 ⁻²⁰

Necessary Materials

Cons. No.	Designation	Part No.
1	Industrial alcohol	Commercially available
2	ATE brake cylinder paste	Commercially available
3	ATE brake fitting spray	Commercially available
4	Perchloroethylene	Commercially available
5	ATE special fluid	Contained in repair kit

MBMANUALS.COM

Sectional View

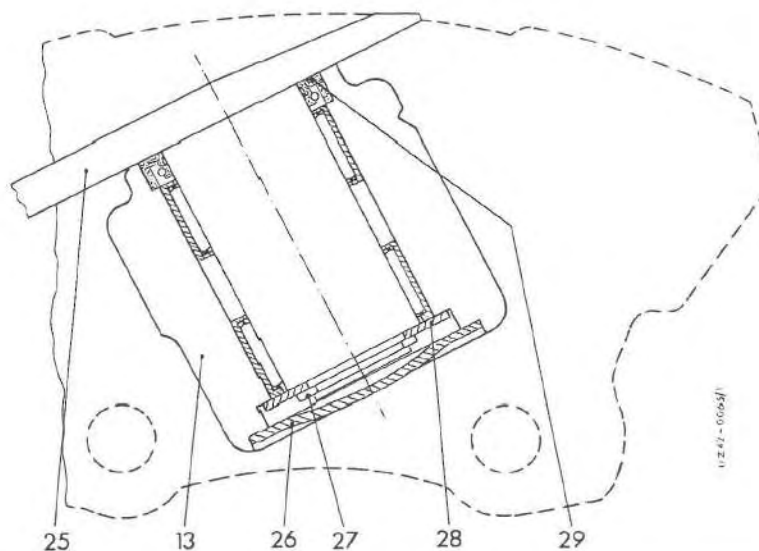


Fixed Caliper

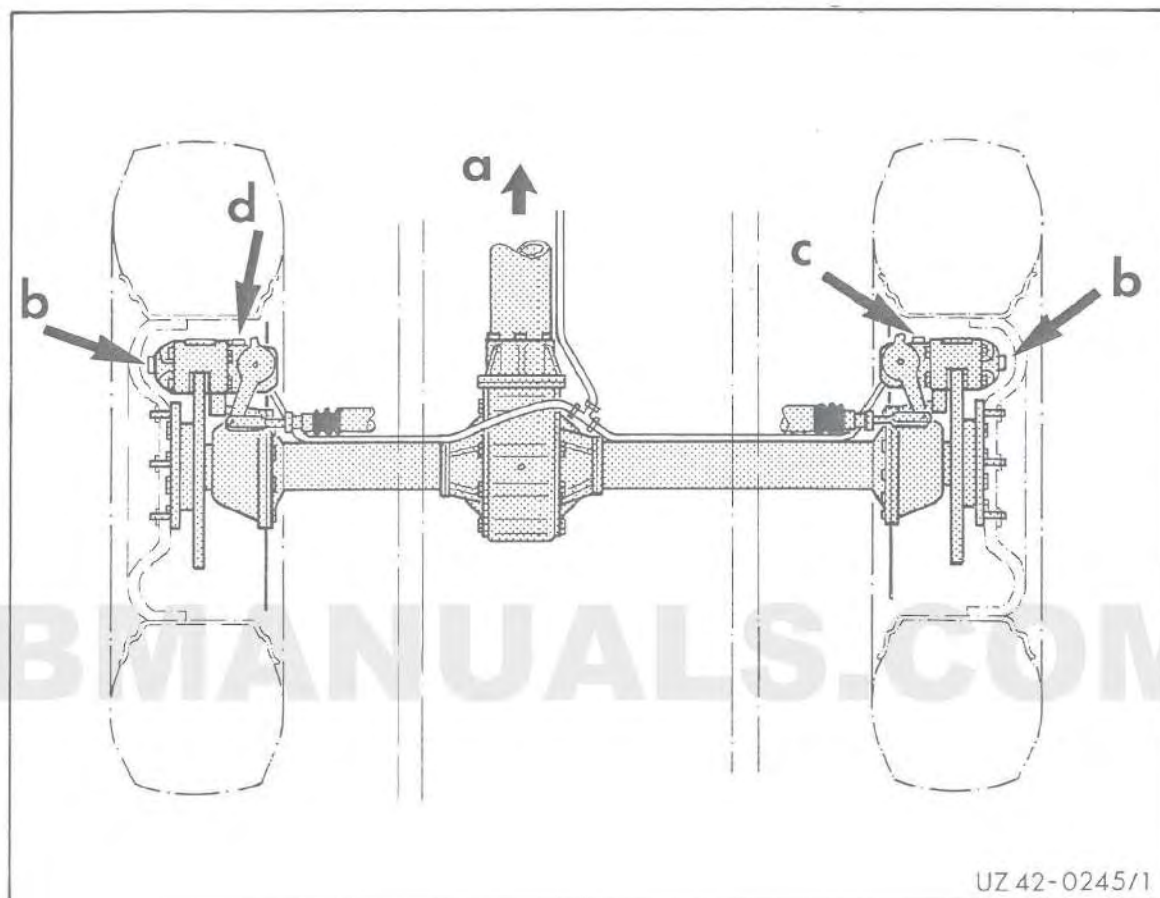
- | | |
|------------------------------|------------------------|
| 1 Hexagon nut (self-locking) | 13 Housing flange |
| 2 Lock washer | 14 Compression spring |
| 3 Housing cover | 15 O-ring |
| 4 O-ring | 16 Supporting ring |
| 5 Adjusting screw | 17 Drive spindle |
| 6 Piston | 18 Screw plug |
| 7 Sealing ring | 19 Copper sealing ring |
| 8 Cover plate | 20 Retaining ring |
| 9 Expander spring | 21 Shim |
| 10 Dust cap | 22 O-ring |
| 11 Spring plate | 23 Restoring shaft |
| 12 Retaining ring | 24 Tapet |

Fixed Caliper

- | |
|--------------------|
| 13 Housing flange |
| 25 Actuating lever |
| 26 End plate |
| 27 Retaining ring |
| 28 Supporting ring |
| 29 Radial seal |



Direction of Rotation Specifications for Adjusting the Combined Fixed Caliper



UZ 42-0245/1

a = Direction of Travel

	b	c	d
Piston to brake disk	counterclockwise	counterclockwise	clockwise
Piston from brake disk	clockwise	clockwise	counterclockwise

Troubleshooting

Hydraulic Brake System

Fault	Cause	Remedy
No or inadequate braking action	Air in the system.	Bleed, top up brake fluid.
	Spring or linkage to ALB modulator broken or detached.	Attach spring, linkage or renew. Adjust ALB according to type plate specifications.
The effect of the brake fades during partial braking.	Leakages in lines or damaged or unuseable sealing rings in master cylinder or brake caliper.	The lines must be sealed. Renew damaged sealing rings.
Warning lamp for brake system lights up in the instrument cluster during driving.	Tandem master brake cylinder faulty.	Check, if necessary, recondition or renew.
	Hydraulic brake circuit I or II failed.	Find leaks in brake system and eliminate.
	Worn brake linings.	Renew brake pads.
	Brake fluid level too low in the reservoir or empty.	Fill reservoir with specified brake fluid up to the upper marking. Check hydraulic brake system, seal and bleed if necessary.
Brake is applied by itself or heats up during driving.	Equalization bores blocked in tandem master brake cylinder.	Clean equalization bores.
	No play at brake pedal.	Adjust pedal play to 10-20 mm.
	Faulty brake valve causing residual pressure in preload cylinder.	Exchange brake valve.
Brake overheats at one axle.	Hydraulic hose jammed or kinked. Valve effect in hose.	Renew hydraulic hose.
Air repeatedly found in the hydraulic brake system.	Breather hole in cap of reservoir clogged. Reservoir or connecting line from the reservoir to the master cylinder leaking at connections.	Clean breather hole in cap. Tighten reservoir or connecting line. Bleed brake system.
	Air is sucked into the system via the brake pressure regulator.	Check brake pressure regulator, renew if necessary.

Troubleshooting

Wheel Brakes

Fault	Cause	Remedy
Brake is not applied and warning lamp for brake system lights up in the instrument cluster during operation.	Air in brake system, leaks in brake system.	Briefly release brake pedal and depress quickly twice during which resistance must be felt. If the brakes still do not apply, stop vehicle by using the handbrake.
Brake is not applied.	IMPORTANT! Stop immediately	
	Brake fluid reservoir empty.	Find leaks in brake system and eliminate. Bleed brake system.
Insufficient brake application or brakes applied unevenly.	Seized brake pads.	Clean clogged shafts in the brake caliper and brake disk. Check dust cap, renew if necessary. Check spring clip and retaining pins, renew if necessary. Install specified brake pads in sets in one axle. Note: Adjust clearance for combined brake caliper.
	Seized piston in brake caliper.	Remove corrosion in brake caliper cylinders and pistons. Release pistons. Renew seals.
	Worn brake pads.	Fit specified brake pads in sets on one axle.
	Brakes have become wet or dirty.	Dry brake by slightly braking when driving and clean. Check brake disk cover.
	Wrong brake pads installed.	Fit specified brake pads in sets on one axle.
	Grooves in surface of brake disk.	Reface brake disks or renew. Observe tolerance specifications of manufacturer.
	Unequal tyre pressure	Correct tyre pressure.
	Oil on brake pads.	Seal wheel hubs or stub axle.

42.11 General

Fault	Cause	Remedy
Brake overheats.	Uneven lining wear.	Fit specified brake pads in sets on one axle. Clean dirty shafts in brake caliper and brake disk. Remove corrosion in brake caliper cylinders and piston, release piston. Check piston return (clearance), release piston if necessary.
	Pressure controlled by ALB too low when brakes fully applied.	Check ALB for correct modulation pressure according to type plate specifications and adjust.
	Closing pressure of pressure reducing valves too low preventing full braking pressure being obtained when fully applying brakes.	Correctly adjust closing pressure of pressure reducing valves.
	Brake pads do not release from brake disks.	Clean dirty shafts in the brake caliper and brake disk. Check dust cap, renew if necessary. Exchange spring clip with retaining pins. Check wheel hub mount, secure if necessary. Check brake disk cover. Remove corrosion in brake caliper cylinders and pistons, release piston.
	Piston return in the brake caliper does not function correctly.	Check piston return (clearance), release piston if necessary. Renew seals.
	Brake valve does not vent completely.	Check function of brake valve. Renew if internal movement is stiff. Check cable ensure cable moves smoothly. Check or adjust pedal play.
	Spring-loaded cylinder does not release fully. Release pressure too low.	Check handbrake valve, renew if necessary.
	Wrong brake pads installed.	Install specified brake pads in sets on one axle.
Brake noisy (squeeking).		

Fault	Cause	Remedy
	Brake linings vitrified. Brake disks scored.	Grind brake pad surface and brake disks, renew if necessary.
		Clean soiled wells in the brake caliper and brake disk. Check dust cap, renew if necessary.
	20° position of piston collar in the brake caliper incorrect.	Check 20° position of piston collar in brake caliper, correct if necessary.
	Shielding plate incorrectly installed.	Install shielding plate correctly in fixed caliper (cut-out facing disk entry).

CAUTION!

Care must be taken to ensure that the close tolerance bolts for mounting the brake caliper on the wheel hub drive housing are always mounted on the disk run-in side.

Check brake disk for vertical and lateral runout as well as thickness tolerance, remachine or renew as required.

The following measures A and B may prove to be of additional assistance on vehicles with the chassis end No. from 435.115 10 002 821
435 117 10 002 756

Measure A: attaching plates (SA 36 267)

1 Remove connecting bolts for the fixed caliper one after the other in the **sequence shown in the diagram** and replace by new fixed caliper connecting bolts (items 1 to 4).

Note: Only use the specified bolts part No. 435 990 72 19.

2 Fit plates (Item 5) **washers**, (Item 6) **nuts** (Item 7) and tighten nuts.

Tightening torque: 34 Nm.

Measure B: installation of close-tolerance bolt in the side of the disk run-in

1 If the close-tolerance bolt is fitted on the disk run- out side (only in vehicles to chassis end No. 084 163), then remove fixed caliper, referring to group 33.

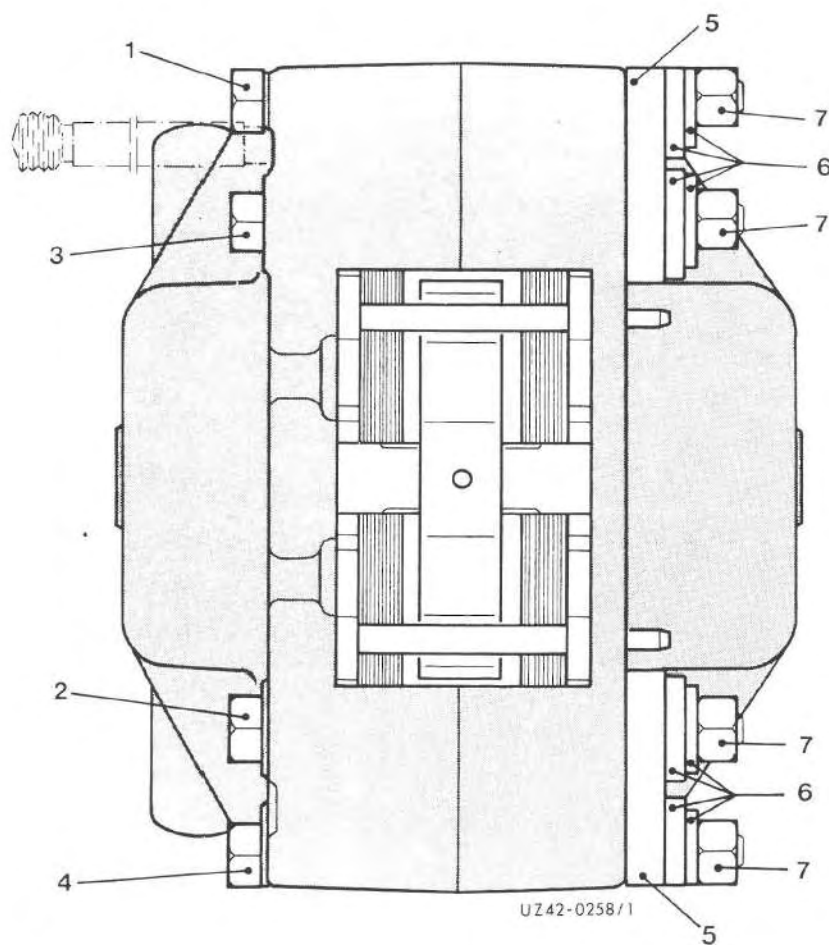
2 Rework the bore on the run-in side on the fixed caliper to $\varnothing, 16.5$ mm H 9.

3 If damaged, the existing hole (disk run-in) of the reduction gear housing should be reworked to $\varnothing, 16.5$ mm H 7.

4 Install close-tolerance bolt on the disk run-in side and tighten.
For tightening torque, see 1.3/1.

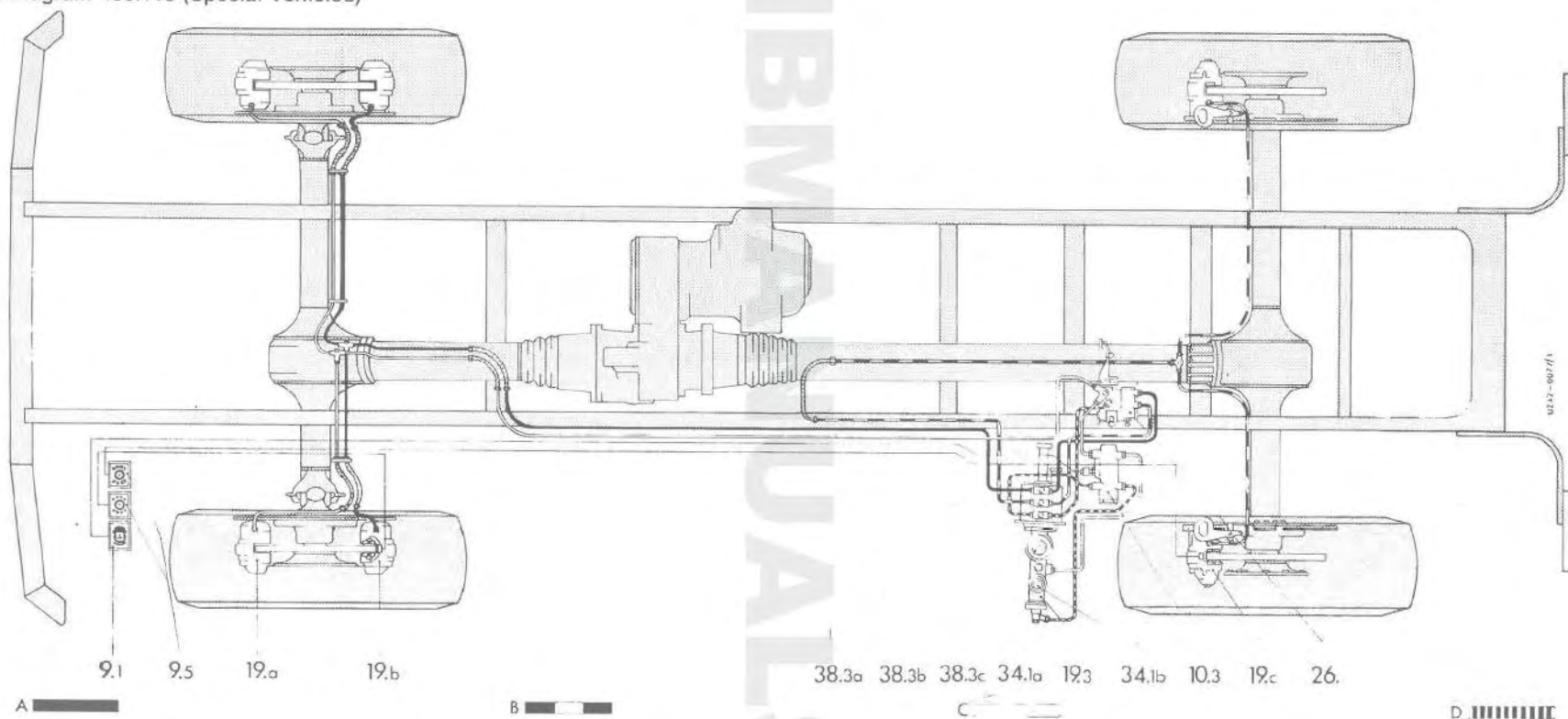
42.11 General

Fault	Cause	Remedy
-------	-------	--------



- 1 – 4 Tightening sequence for fixed caliper connecting bolts (tightening torque 58– 2 Nm)
- 5 Plates
- 6 Washers
- 7 Nuts (tightening torque 34 Nm)

Brake Diagram 435.115 (Special Vehicles)

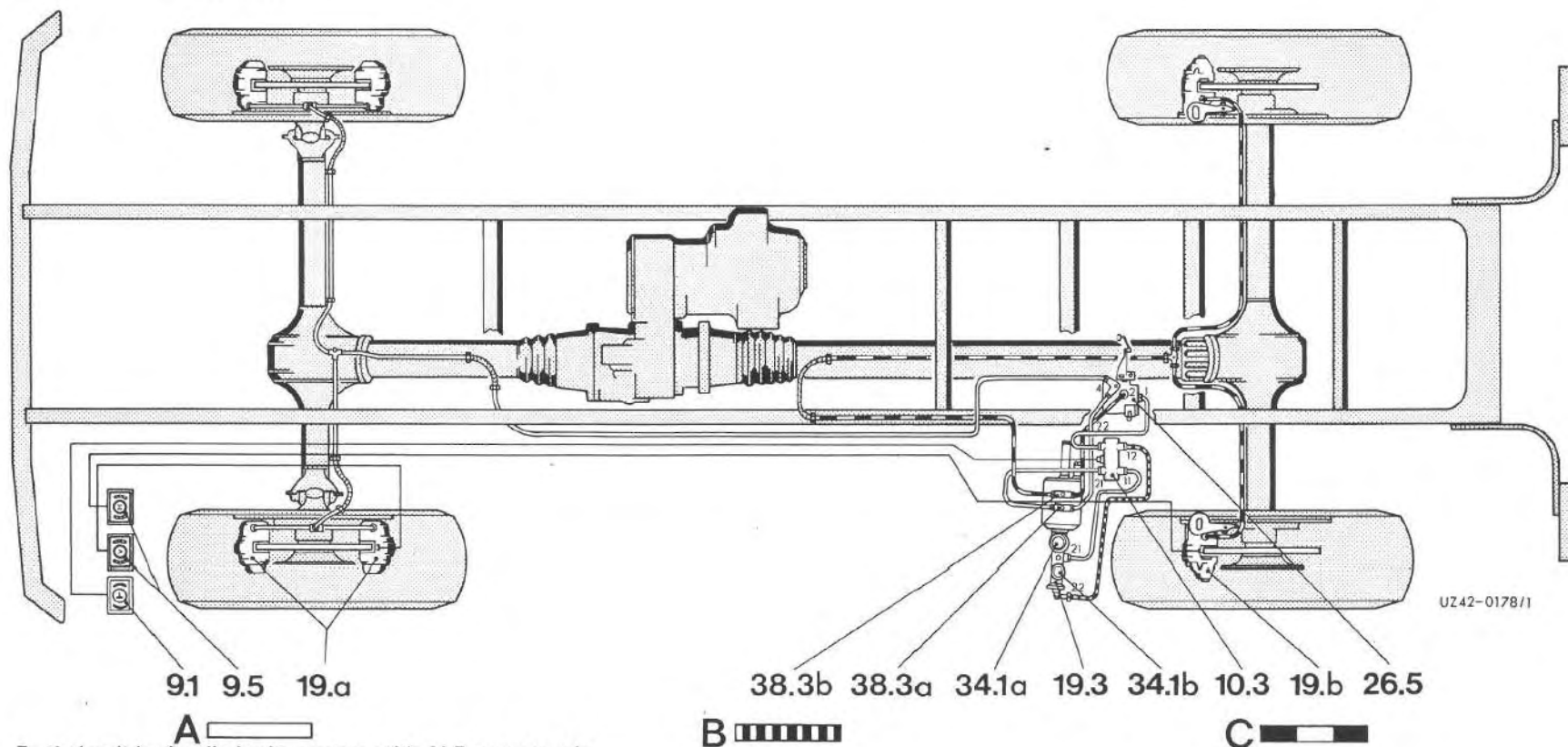


Dual circuit hydraulic brake System with ALB on rear axle and 1/2 front axle
to chassis No. 435.115-084 388

- | | | | |
|------|---|-------|--|
| A | Circuit 1 modulated, front axle | C | Circuit 1 unmodulated, front axle |
| B | Circuit 2 modulated, rear axle | D | Circuit 2 unmodulated up to ALB modulator |
| 9.1 | Warning lamp, pressure difference (SA 35 847) | 19.3 | Master brake cylinder (dual circuit) |
| 9.5 | Warning lamp, brake wear indicator | 26. | Tandem modulator (ALB modulator) |
| 10.3 | Electrical warning switch for pressure difference (SA 35 847) | 34.1a | Hydraulic tank circuit 1 |
| 19.a | Fixed caliper, circuit 1 unmodulated | 34.1b | Hydraulic tank circuit 2 |
| 19.b | Fixed caliper, circuit 2 modulated | 38.3a | Test connection 1/2 front axle, brake circuit 1, modulated |
| 19.c | Fixed combined caliper, service and parking brake circuit 2 modulated | 38.3b | Test connection rear axle brake circuit 2, modulated |
| | | 38.3c | Test connection brake circuit 2nd rear axle, unmodulated |

See 43.11-1.5/1 to 1.5/28 for symbols.

Brake Diagram 435.115/117



Dual-circuit hydraulic brake system with ALB on rear axle
from chassis No. 435.115-084 389/435.117 - 071 796

- | | | |
|-----|---|-------------|
| A | Front axle | (circuit 1) |
| B | Rear axle unmodulated
before ALB | |
| C | Rear axle modulated
after ALB | (circuit 2) |
| 9.1 | Pressure difference warning lamp
(SA 35 847) | |
| 9.5 | Brake lining wear indicator
front axle/rear axle | |

- | | |
|-------|--|
| 10.3 | Pressure difference warning switch (SA 35 847) |
| 19.a | Fixed caliper unmodulated circuit 1 |
| 19.b | Fixed combined caliper modulated circuit 2 |
| 19.3 | Master brake cylinder dual-circuit |
| 26.5 | Single-circuit ALB modulator |
| 34.1a | Equalizing tank circuit 1 |
| 34.1b | Equalizing tank circuit 2 |
| 28.3a | Test connection brake circuit 1 unmodulated |
| 38.3b | Test connection brake circuit 2 modulated |

See 43.11 - 1.5/1 to 1.2/28 for symbols.